

CONNECTING THE DOTS

**APPLICATIONS OF
GENERATIVE AI
ACROSS INVESTMENT
MANAGEMENT
RESEARCH**



TABLE OF CONTENTS

1. GENERATIVE AI IS REDEFINING THE PROCESS OF FINANCIAL RESEARCH	1
2. A FRAMEWORK APPLICATIONS OF GENERATIVE AI IN INVESTMENT MANAGEMENT	4
3. CONSIDERATIONS FOR AI ADOPTION IN INVESTMENT MANAGEMENT	7
ABOUT BRIGHTWAVE	13

1. GENERATIVE AI IS REDEFINING THE PROCESS OF FINANCIAL RESEARCH

Generative AI has seen unprecedented adoption by firms in every sector of the economy, and in financial services the value proposition is especially clear - whether creating workflow efficiencies or driving insights at scale, generative AI is already changing the way finance professionals work and do business. According to a recent KPMG survey of executives leading organizations across all industries with revenues of \$1 billion or more¹, 83% expected generative AI investments to increase over the next 3 years, while 78% expressed confidence in seeing returns on their generative AI investments over the next 1 to 3 years. In another study, the McKinsey Global Institute estimated that generative AI's impact on productivity could add \$2.6 - \$4.4 trillion annually across the global economy, increasing AI's current impact by 15-40%². In other words, if you aren't developing or implementing an AI strategy, you're almost certainly falling behind your peers.

Within the investment management industry many workflows are well-positioned to benefit from the advances of generative AI, principally with respect to process efficiency, comprehensiveness of analysis, and improved decision making in support of better returns. For organizations in both public and private markets the heart of a good position lies in the development of a robust thesis, and as anyone who's done the work knows, excellent thesis development can be time-consuming and intellectually taxing, but the way this manifests varies by investing style. For example, a smaller value-based investment firm with a lean team of generalists may constantly come up against resource constraints. On the other end of the spectrum, a multi-manager model with hyper-sectorized coverage may face different challenges around information overload and relevance latency.

¹ KPMG. (2024). (rep.). GenAI Survey - 2024. Retrieved from <https://kpmg.com/kpmg-us/content/dam/kpmg/pdf/2024/kpmg-genai-survey-august-2024.pdf>.

² Chui, M., Hazan, E., Roberts, R., Singla, A., Smaje, K., Sukharevsky, A., Yee, L., & Zimmel, R. (2023, June 14). McKinsey digital.

Collating and analyzing data from 100+ hours of user research interviews with both sell-side and buy-side organizations, we've identified several representative use cases where an AI research platform can add value for firms in the financial services sector:

- **Fact-finding & question answering:** Firms of all investment types expect accurate, comprehensive responses to open-ended questions on markets & sectors, breaking news, financial metrics and more. This is a high-volume use case that drives significant time savings owing to the frequency with which the need arises.
- **Summarization:** From earnings call transcripts to CIMs, at-a-glance summaries of long-form multimodal documents are useful and widely available. In order for summaries to be useful, they must extract information with consistently high quality, a feature that differentiates Brightwave from competitors. Similarly, question-answering functionalities must also identify responses with a high degree of accuracy. Both of these use cases create meaningful efficiencies, with demand volume for summarization technology tied to events like earnings season, portfolio turnover, or new deal diligence.
- **Primary Research:** Finance professionals getting up to speed on new sectors, names and subjects are increasingly turning to generative AI solutions to accelerate the research process on unfamiliar topics. Synthesis capabilities, or the ability to organize fact patterns across multiple documents into a coherent narrative that supports a decision or insight, is the defining capability for solutions in this class.
- **Benchmarking:** Whether across deals in private market transactions or among sector peers in the public markets, a common use case involves the ability to evaluate many companies simultaneously across dozens of dimensions. With generative AI, analysts can accomplish in a matter of minutes a common, detail-oriented task that could easily take hours using traditional methods.

- **Change Analysis:** Contextualizing historical trends (e.g. identifying changes in management tone or guidance across quarters) is a task that is exceptionally well-suited to the capabilities of generative AI systems. In this use case particularly we see teams seeking out solutions that allow for nuanced assessment of specific factors unique to each firm's mandate, strategy, or individual investment opportunity. Because generative AI enables investment professionals to take a magnifying glass to huge volumes of content, accelerating efficiencies can create meaningful opportunities for alpha generation as the marginal cost of fine-grained analysis approaches zero.
- **Semantic Screening:** The ability to filter and sort entities based on non-market-data factors is perhaps best illustrated with an example: "Identify mine operators with significant regulatory risk related to wastewater treatment compliance, where non-compliance could result in significant fines or operational disruptions." While less frequent than question answering, this capability enhances idea generation and enables the adaptation of strategies to new markets and entities.
- **Idea Generation:** Finally, generative AI has enabled an increase in the volume and quality of idea production. Crucially, this activity is not something generative AI systems can perform autonomously, so training and enablement are critical to extracting maximum value. What we hear from firms across the industry is a desire for tools that enable analysts to consider the totality of factors relevant to an investment decision, leading to a greater number of higher-conviction ideas in the same period of time and with lower total effort.

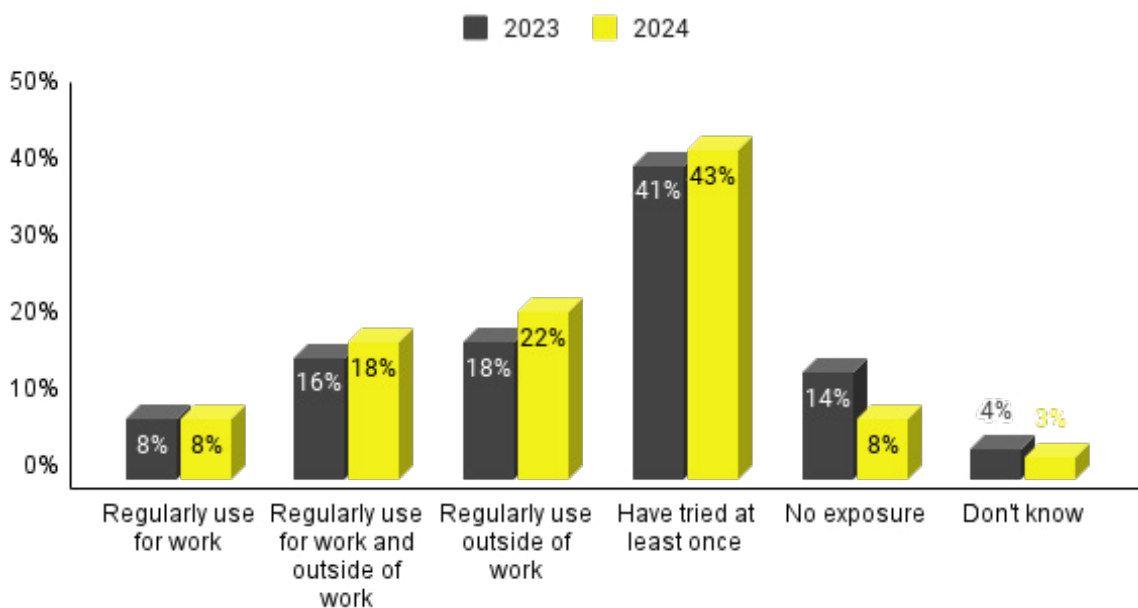
Our research also suggests that experience level is a key determinant of the value finance professionals look to generative AI tools to provide, with more junior analysts and early career professionals tending to appreciate the efficiency and comprehensiveness the technology brings to high-volume, time-sensitive workloads, while more seasoned individuals use the technology to surface non-obvious relationships, understand and contextualize sector-level trends, generate ideas, and harden and refine investment theses.

These examples and findings are non-exhaustive, and characterize just a sampling of the opportunities presented by generative AI. Fundamentally, language models represent a phase change in humans' ability to process and understand large volumes of information, and nowhere are they more relevant than the financial services sector.

2. A FRAMEWORK APPLICATIONS OF GENERATIVE AI IN INVESTMENT MANAGEMENT

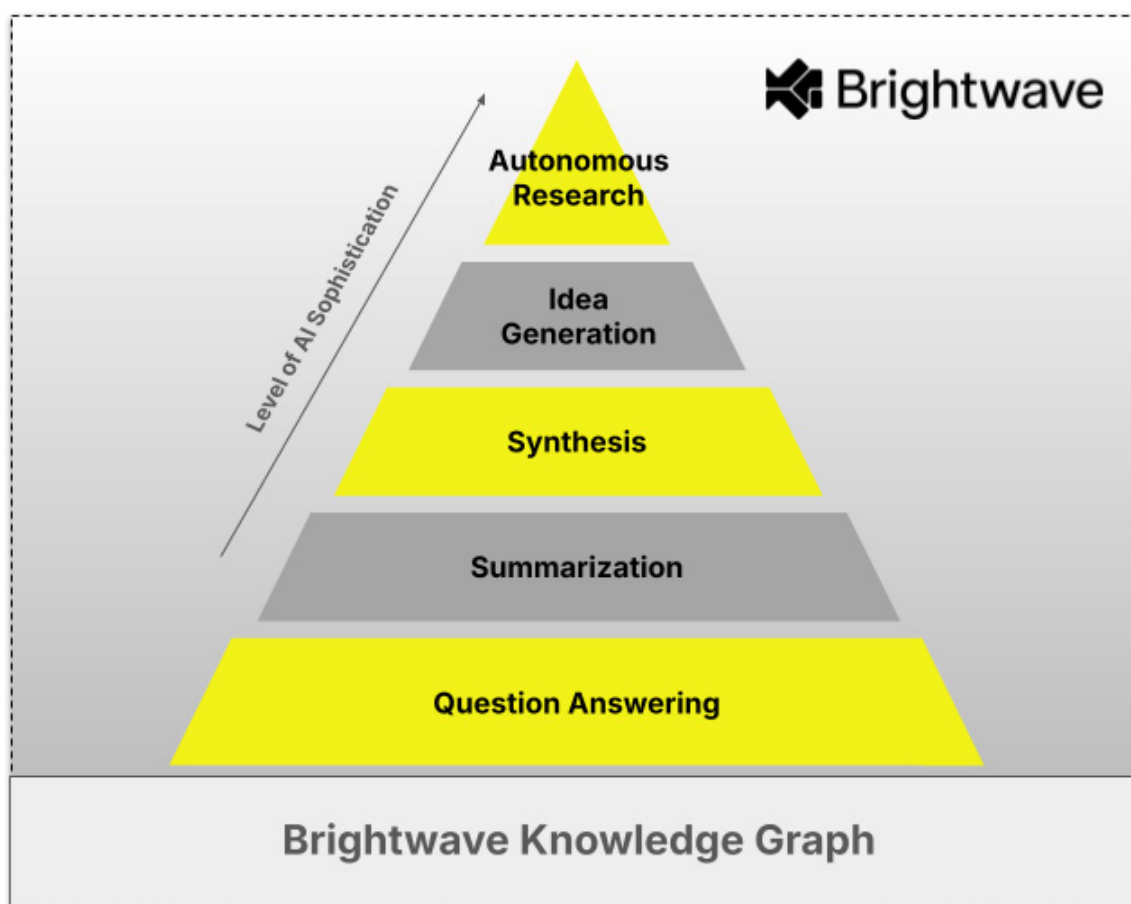
Interest among financial services professionals in the practical applications of generative AI is growing. According to a 2024 survey conducted by the McKinsey Global Institute, 48% of financial services professionals regularly use generative AI either at or outside of work, a +6% increase from those surveyed in 2023. Notably, the share of professionals who had no exposure to generative AI dropped by 43% in the last year, suggesting that AI is rapidly permeating multiple aspects of personal and professional life.

Generative AI Use in Financial Services, 2023-2024



As outlined above, generative AI has a role to play in many different aspects of the investing and research workflow. We conceptualize the relationships between tasks in terms of a Maslow's hierarchy of needs framework, with basic, objective ground-truth fact at the base and autonomous, open-ended agentic research at the top. The pyramid represents a scaling of both value and technical difficulty, as higher levels represent systems capable of complex reasoning closer to what we would expect from a human expert.

Each level of the pyramid can also be thought of as mapping to specific investment research workflows. Analysts, for example, are often tasked with creating or updating financial models using financial metrics derived from publicly available SEC filings or proprietary company financials. At this question-answering or summarization level, AI's utility lies in its ability to quickly search across vast amounts of information, identify the correct answer, and present it in a way that is immediately useful to a concrete workflow or output (e.g. updating a financial model or filling out an investment committee template).



As we move up the pyramid into synthesis and idea generation, generative outputs become more subjective, complex, and insightful. Here especially the limitations of chat as a form factor become more pronounced, and data product-style user experiences tend to dominate. When trying to create value at these levels of the hierarchy, it is crucial that organizations take a thoughtful approach and select configurable tools that work within the context of their specific investment research and workflow needs.

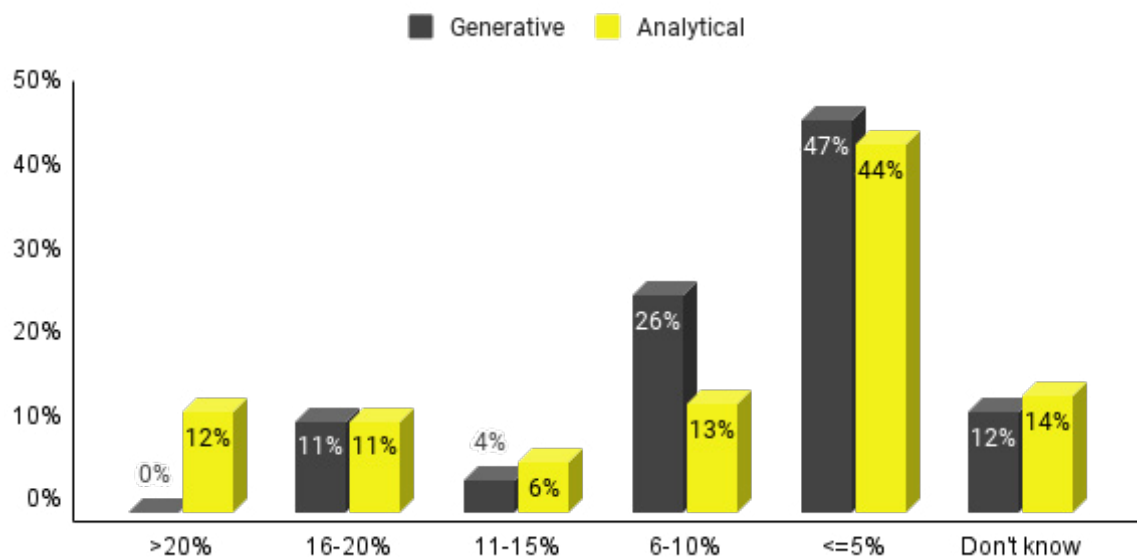
The current state of the art for vertical-specific generative AI systems, embodied by the approach we take at Brightwave, does not involve training a single monolithic model that is responsible for all tasks the system performs. Instead, superior results follow from the interoperation of many individual AI systems, each of which is purpose-built for specific tasks and optimized for accuracy and task-specific performance. Examples of these tasks can include search and information retrieval, intent understanding, research planning and execution, or producing and critiquing generative outputs. The challenge of building these individual AI subsystems is itself a significant machine learning and engineering lift, but optimizing multiple separate systems to work together in service of a reliably high-quality output is an even greater challenge, leading many teams to consider procuring third party solutions rather than building in-house.

As in many other domains, the quality of the data available to generative AI systems is a key determinant of overall performance. The most powerful generative AI systems offer high-signal primary sources such as first party offerings, integrate deeply with internal databases, and leverage proprietary data sets to ensure that all relevant factors are available for consideration. At Brightwave, we are developing a proprietary knowledge graph that describes a high-resolution map of the global economy in unprecedented detail, and this powerful asset underpins and enhances all aspects of the reasoning and research capabilities of the Brightwave service.

3. CONSIDERATIONS FOR AI ADOPTION IN INVESTMENT MANAGEMENT

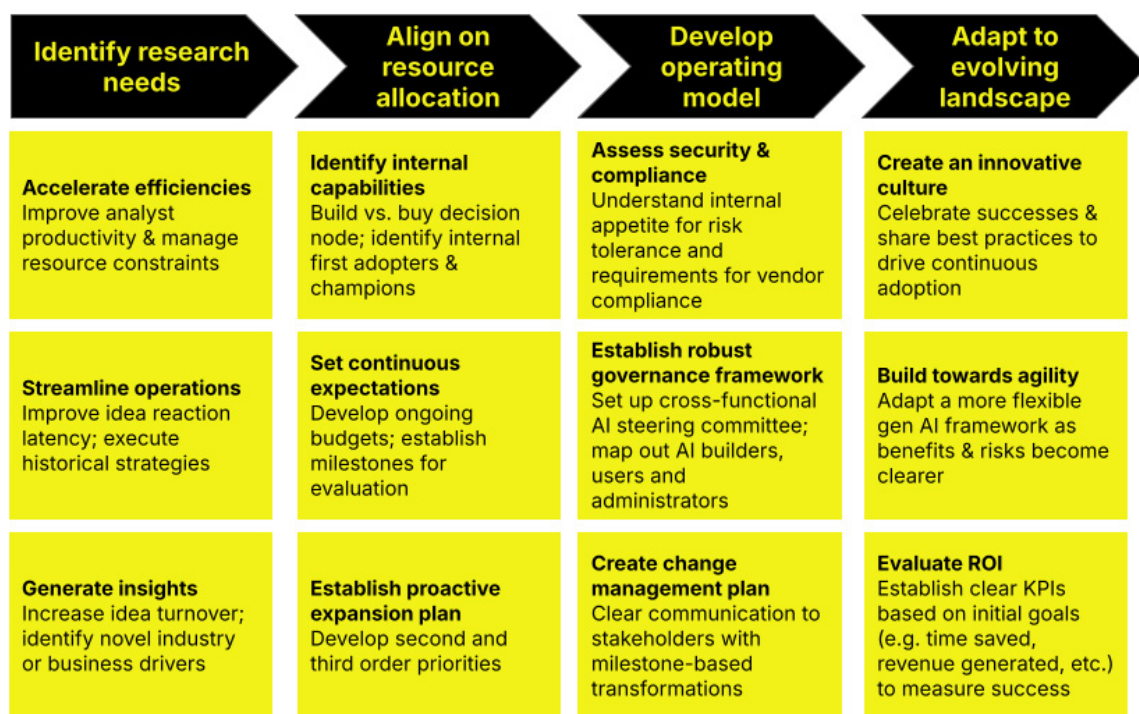
As generative AI tools continue to evolve, it is becoming increasingly clear that many financial services firms are willing to put real money on the table in order to avoid falling behind. Considering that these technologies are still in their infancy, it is impressive that nearly half of financial firms (~41%) are already spending >5% of their digital budgets on generative AI³. Nevertheless, with idea generation driving top-line growth and workflow efficiencies supporting cost control, there remains substantial upside for investment in all aspects of AI awareness, education, and adoption.

Share of organization's digital budget spent on generative vs. analytical AI



³Chui, M., Hazan, E., Roberts, R., Singla, A., Smaje, K., Sukharevsky, A., Yee, L., & Zemmell, R. (2023, June 14). McKinsey digital.

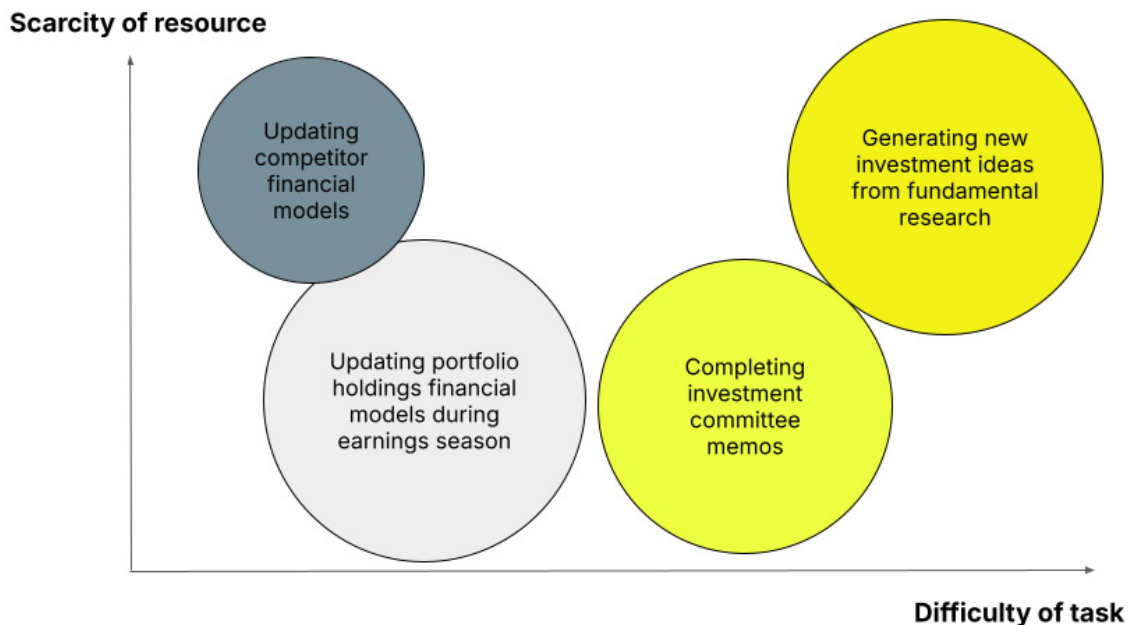
When considering a tactical framework for operationalizing generative AI, organizations should begin from a starting point that considers their individual research needs and outputs. From there, understanding resource availability and allocation (both human and capital) can inform the development of a bespoke operating model tailored to each business. Change management, arguably the most difficult and underrated component of a business transformation, will only be successful if clear owners and milestones are initially designated. Finally, the conversation around generative AI tools should be an ongoing dialogue as the risk-benefit analysis of existing and emergent technologies reshapes the technological landscape.



STEP 1: IDENTIFY RESEARCH NEEDS

While it may be tempting to build, buy or partner on generative AI tools based on the current hype of what is technologically possible, ensuring long-term success should be predicated first and foremost on the needs of the business and concrete workflow outputs.

One potential framework for assessing research needs is to consider the parallel dimensions of resource scarcity and task difficulty. For example, a long-only value investor focused on public equities might prioritize technological efficiencies for key workstreams using the below graph. They might then decide that conforming to a “low-hanging fruit” strategy is the appropriate first step for their business, searching for AI tools that expedite financial modeling extraction across publicly filed documents.



In most cases, trying to solve for every workstream is counterproductive (not to mention expensive). While generative AI can certainly bring incredible efficiencies to the investment research process, it should not be treated as a panacea for all resource gaps.

STEP 2: ALIGN ON RESOURCE ALLOCATION

Vendor agreements in the generative AI space are developing alongside the technology itself, offering buyers an opportunity to customize their purchases in a nascent market. According to a recent KPMG survey⁴, the majority of organizations are seeking outside help: 50% are buying or leasing the technology directly from vendors; 29% are experimenting with a mix of building, buying, and partnering; and only 12% are choosing to build in-house. For most organizations, the build vs. buy vs. partner decision is contingent both on the complexity of their research needs and the ability to attract the requisite engineering and machine learning talent. The costs associated with building an internal AI model comparable to market offerings may be prohibitively high in sectors that have not historically attracted cutting-edge engineering talent.

Setting continuous expectations and establishing a roadmap for AI capabilities can help organizations arrive at a concrete plan for AI spending and expected ROI. In particular, aligning on milestones and key unlocks for additional spending can provide clarity in a rapidly-shifting market by centering organizational priorities around their own research needs.

⁴KPMG. (2024). (rep.). GenAI Survey - 2024. Retrieved from <https://kpmg.com/kpmg-us/content/dam/kpmg/pdf/2024/kpmg-genai-survey-august-2024.pdf>.

STEP 3: DEVELOP OPERATING MODEL

Realizing true top-line or productivity goals from any technology is predicated on the successful adoption of change management strategies. Within the highly regulated, competitive industry of investment management, certain considerations play highly important roles. One particularly important consideration is security and compliance. These requirements range from data privacy and protection to federally-mandated disclosures of holdings, and any AI tool must prove that its foundational structure is compatible with finance-specific regulations.

From there, establishing a robust governance structure to research, evaluate and implement any new generative AI technology is critical to success. Generative AI steering committees comprised of cross-functional stakeholders who can each represent different viewpoints in the business are highly beneficial to uncovering the most valuable common denominators. Collectively, the committee should be responsible for all aspects of end-to-end AI transformation, including key technological architecture considerations, integration with existing tech stack, and partnerships with end users.

Steering committees are helpful in establishing a single point of ownership that can act decisively when needed, especially in situations where a quickly-changing landscape like generative AI can muddy the waters on decision making. However, these committees must not act in a vacuum: clear communication and buy-in from the rest of the business should be an equally important mandate. Setting any new technology up for success requires a serious change management effort that should not be underestimated.

STEP 4: ADAPT TO EVOLVING LANDSCAPE

One thing that has been made abundantly clear in the first phase of generative AI adoption is that the impact and fast pace of technology development will continue to accelerate innovation in the coming months and years. Any kind of generative AI implementation will not be a “one-and-done” project; success requires an evolving plan that considers the impact of future technology acceleration and regulatory developments.

One way in which organizations can set themselves up for success is to invest resources in building an innovative, AI-first culture where groups regularly share best practices, celebrate wins, and provide feedback on new tools. These investments set the stage for increased agility as AI tools develop, ensuring that the ultimate ROI of any AI investments remains positive for the business.



ABOUT BRIGHTWAVE

Brightwave is an AI financial research platform that delivers actionable insights for teams that help move the needle on investments. Brightwave's purpose-built AI systems synthesize insights across an expansive universe of content, enabling financial professionals to accelerate research workflows, improve investment theses, and step into every interaction prepared to lead with a sophisticated view. Founded by engineers with deep expertise in AI and finance infrastructure, Brightwave's leadership team has decades of experience building systems responsible for handling the world's most sensitive and mission critical data. Brightwave's investors include Decibel, OMERS Ventures, Point72 Ventures, Moonfire and angels from OpenAI, Databricks, Uber and LinkedIn.

For more information, visit our website at <https://www.brightwave.io/>.